CERTIFICATE

This is to certify that the project synopsis entitled **"Agentic AI-Driven Hybrid Digital Forensics & Cyber Threat Intelligence Framework"** is being submitted by **Shivansh Chauhan (230211337), Sofia Ali (23022564), Sahil Yadav (23021810), and Krish Yadav (230111120)** for the partial fulfilment of the requirement for the award of the degree of **B. Tech - CSE** in the Department of Computer Science & Engineering, Graphic Era Deemed to be University, Dehradun, and is a record of their own work carried out under my supervision. The matter embodied in this synopsis has not been submitted in any other university or institution for the award of any degree.

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DECLARATION

We hereby declare that the project synopsis entitled **"Agentic AI-Driven Hybrid Digital Forensics & Cyber Threat Intelligence Framework"** is being submitted for the partial fulfilment of the requirement for the award of the degree of B. Tech - CSE in the Department of Computer Science & Engineering, Graphic Era Deemed to be University, Dehradun, is our original work and has not been submitted to any other university or institution for the award of any degree or diploma.

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ABSTRACT

Security teams today are flooded with authentication logs, system events, and network-flow data, and a purely rule-based monitoring setup usually can't keep up with attack behaviour it hasn't seen before — manually correlating everything into an actual investigation is even slower. This synopsis proposes a Digital Forensics and Cyber Threat Intelligence (DFIR-CTI) framework that brings together hybrid machine-learning detection, rule-based detection, forensic evidence correlation, IOC extraction, MITRE ATT&CK mapping, and a single agentic AI layer that plans the investigation, writes the report, and suggests mitigation steps — all inside one dashboard.

At the core of the detection engine is a hybrid setup: Isolation Forest gives a fast, unsupervised anomaly score as a baseline; a supervised XGBoost model is trained to classify known attack categories using MITRE ATT&CK-labelled data; and a deep autoencoder catches unknown or zero-day-type behaviour by looking at reconstruction error. These three scores are fused and handed to a single agentic AI layer, which looks at the correlated alerts, forensic evidence, and threat-intel context, plans out an investigation, drafts an incident report, and recommends concrete steps to contain threats like ransomware, malware, and phishing — all inside a safety-bounded, human-in-the-loop lab setup.

The system will be tested on the UWF-ZeekData24 dataset (2025), a fairly recent, enterprise-scale dataset labelled against MITRE ATT&CK, along with a controlled lab scenario of our own. Precision, recall, F1-score, and false-positive rate are used to judge the detection engine, while the agentic layer is judged separately on task completion, plan quality, and how well-grounded its output is in the actual evidence. Along with the working prototype, we also want to find out how much a hybrid detection ensemble plus a single autonomous agent can realistically help a human analyst — something we hope to write up as a short research paper alongside the software.

1. INTRODUCTION

1.1 Background

Cyberattacks leave behind a lot of data — system logs, authentication events, network traffic, application logs — and traditional rule-based monitoring is good at catching known attack patterns but tends to miss anything unusual it wasn't written to detect. Even when an alert does fire correctly, someone still has to pull the evidence together, judge how serious it is, and decide what to do next. Digital forensics is what lets an investigator reconstruct what actually happened after an incident, cyber threat intelligence (CTI) turns raw technical observations into something a team can act on, and lately agentic AI — AI systems built on large language models that can use tools and act with some autonomy — is being tried out for exactly this kind of triage, investigation, and response work.

That's essentially what this project is trying to build: a framework that ties together digital forensics, threat intelligence, rule-based detection, hybrid ML detection, and a single agentic AI component. The platform ingests security data, flags suspicious behaviour, correlates related events, pulls out indicators of compromise, maps whatever it finds to MITRE ATT&CK, and then passes the whole correlated case to the agentic layer, which plans the investigation, writes up the report, and suggests how to deal with threats like ransomware, malware, and phishing — with everything visible through one dashboard.

1.2 Scope of the Project

The technical work here has two main parts. The first is a hybrid detection module — an unsupervised Isolation Forest as the baseline, a supervised XGBoost classifier for attacks we already have labels for, and a deep autoencoder to catch behaviour that doesn't match anything known — whose combined output feeds both the rule engine and the agent. The second part is the agentic AI layer itself, deliberately kept to a single orchestrating agent instead of a multi-agent setup, mainly because that is what is realistic to build and evaluate properly within a 12-week timeline. The intent is a complete framework, not just another detection demo. That said, this is still a student-scale prototype rather than a commercial SIEM or SOAR product — every experiment, including anything the agent recommends or triggers, stays inside public datasets or our own controlled, isolated lab.

2. LITERATURE REVIEW

A lot of intrusion-detection research leans on a handful of benchmark datasets. Sharafaldin et al. built CIC-IDS2017 by generating five days of realistic background traffic mixed with common attacks — brute force, DoS, web attacks, infiltration, botnets, port scans — and it is still one of the most-used benchmarks around [1]. The problem is that it is from 2017. More recently, Elam et al. put out UWF-ZeekData24, an enterprise-scale dataset built from Zeek logs and labelled against the MITRE ATT&CK framework using controlled mission logs, which fixes a lot of the labelling and realism problems older, crowd-sourced datasets have [7]. We have picked this as our main dataset for that reason — how current the traffic is matters a lot if the model is supposed to reflect what attacks actually look like today.

Isolation Forest (Liu, Ting, and Zhou) works by recursively partitioning the feature space; since anomalies are rare and different, they get isolated in fewer splits than normal points, which keeps the algorithm cheap compared to distance- or density-based detectors [2]. That is exactly why it makes a good baseline for a tabular, student-scale dataset. Zahoor et al. compared it against One-Class SVM on TON_IoT and found One-Class SVM actually did better on precision and recall in their setup, but Isolation Forest still held its own as a lightweight, interpretable baseline once feature-importance analysis and LIME-based explanations were added [3]. That is more or less the role we are giving it here too — a cheap first pass, not the only detector.

To go beyond one model, some recent work pairs gradient boosting with deep representation learning. Abdulganiyu et al.'s XIDINTFL-VAE uses a class-wise focal-loss variational autoencoder alongside XGBoost specifically to deal with class imbalance in network traffic, and reports noticeably better detection of minority-class attacks than single-model baselines [8]. That is basically the reasoning behind our hybrid setup: XGBoost handles the attack categories we already have labels for, the autoencoder's reconstruction error catches whatever is rare or unlabelled, and Isolation Forest stays in as a cheap cross-check.

On the forensics side, Al Balushi, Shaker, and Kumar reviewed how ML gets used across digital forensic investigations and found that different algorithms tend to get used for different sub-tasks — timeline reconstruction, evidence correlation, artefact classification — without much integration between them into one workflow [4]. Jeet, Karn, and Kumar's hybrid TDA + LSTM model, tested on CIC-IDS2017, makes a similar point from the detection side: a plain Isolation Forest baseline gets noticeably better once it is paired with complementary techniques or temporal context [5]. Both point the same direction — towards a fused, multi-model engine rather than one algorithm doing everything.

There is also a growing amount of work on agentic AI for security operations specifically. Ismail et al. describe a hyper-automation approach where an agentic AI orchestrates SOAR-style tasks, arguing this can take a lot of the triage, contextual investigation, and playbook-execution load off human analysts instead of leaving it to static runbooks [9]. Our agentic layer follows that idea but stays deliberately narrower — one orchestrating agent covering report analysis, investigation planning, report generation, and mitigation recommendation for ransomware, malware, and phishing — rather than a full multi-agent SOC, which the literature suggests gets hard to coordinate and even harder to evaluate properly at this scale.

Finally, MITRE ATT&CK gives a shared, well-maintained vocabulary for describing adversary tactics and techniques, and it has become close to a standard in both industry tooling and academic threat-intelligence work [6]. Mapping our detections and alerts onto ATT&CK techniques means both the dashboard and the agent can talk about a case in terms security teams already understand, instead of just producing a number.

Put together, this leaves a few gaps worth addressing: most student-scale systems test one detection model at a time instead of a fused ensemble, they are rarely evaluated on genuinely current, ATT&CK-labelled traffic, and very few

DFIR-CTI prototypes actually connect their detection output to an agent that can plan an investigation and suggest mitigation rather than stopping at a dashboard. That is the gap this framework is aimed at closing.

3. PROBLEM STATEMENT & OBJECTIVES

3.1 Problem Statement

A lot of academic cybersecurity projects focus on prevention or on one detection task in isolation, and stop once they have produced a score or an alert. In a real environment, though, a security team also needs to know what to do with that alert — which host is affected, what actually happened, and how fast they need to move, especially against something like ransomware or phishing. This project is trying to address a few specific problems:

- Security logs are heterogeneous, noisy, and difficult to analyse manually.
- A single detection model — rule-based or a lone ML algorithm — tends to miss either known attacks it was not tuned for, or unknown behaviour it has never seen.
- Many benchmark datasets used in student projects are several years old and no longer reflect current attack traffic.
- Forensic evidence, alerts, and threat-intelligence findings usually sit in disconnected tools, and even once correlated, still need a human to plan the investigation and write the report.
- Investigators need a fast, evidence-backed recommendation for containing common threats such as ransomware, malware, and phishing, not just a pile of raw alerts.

The idea here is to close that gap by pairing a fused, hybrid detection ensemble with a single agentic AI layer that plans the investigation, drafts the report, and recommends mitigation — so the whole thing works as one connected workflow rather than a set of tools bolted together.

3.2 Objectives

- Design and implement a unified, complete framework for collecting, processing, correlating, investigating, and reporting on cybersecurity events and forensic artefacts.
- Develop a hybrid detection engine combining Isolation Forest (unsupervised baseline), XGBoost (supervised known-attack classification), and a deep autoencoder (unknown/zero-day anomaly detection), with a documented score-fusion strategy.
- Apply feature engineering to convert raw security logs or network-flow records into model-ready data, using the recent UWF-ZeekData24 dataset as the primary source, with a self-generated lab dataset as a fallback if pre-processing or labelling proves impractical within the timeline.
- Build a rule-based detection engine for known suspicious patterns such as brute-force attempts, unusual login behaviour, or abnormal network activity.
- Extract and structure indicators of compromise, including hashes, IP addresses, domains, filenames, and other relevant artefacts, and map observed behaviours to appropriate MITRE ATT&CK techniques.
- Correlate ML detections, rule-based alerts, forensic evidence, and threat intelligence into incident cases.
- Design and implement a single agentic AI layer that performs report/log analysis, investigation planning, automated report generation, and mitigation recommendation for ransomware, malware, and phishing, operating with human-in-the-loop confirmation inside the isolated lab environment.
- Evaluate the hybrid detection engine and the agentic layer separately, using suitable quantitative and qualitative metrics, and document limitations, false positives, and any agent-reasoning errors.
- Develop a dashboard and agent-generated incident-response report that communicate the attack timeline, evidence, and recommended mitigation.

4. PROPOSED METHODOLOGY

4.1 System Architecture

The system follows a modular design. Existing forensic and analysis tools can be used to generate or inspect evidence, but the software we are actually building handles ingestion, normalisation, hybrid ML detection, correlation, agent reasoning, visualisation, and reporting. Compared to the earlier version of this project, the main structural change is the Agentic AI Layer — the pipeline no longer stops at a dashboard, it ends with a single agent that takes the correlated case and turns it into an investigation plan, a report, and a mitigation recommendation.

Layer	Main Responsibilities
Data Sources	Windows/Linux logs, authentication events, network-flow data (UWF-ZeekData24), forensic artefacts, malware-analysis outputs
Preprocessing	Parsing, cleaning, normalisation, timestamp handling, deduplication, and feature engineering
Hybrid Detection Engine	Isolation Forest (baseline), XGBoost (known-attack classification), deep autoencoder (unknown-behaviour detection), and score fusion
Rule Engine	Known-pattern detection, thresholds, signatures, and deterministic alerts
Correlation Layer	Combining alerts, anomalies, IOCs, hosts, users, timestamps, and forensic evidence into cases
Threat Intelligence	IOC enrichment, ATT&CK mapping, and structured intelligence records
Agentic AI Layer	Case reasoning, investigation planning, automated report generation, and mitigation recommendation for ransomware, malware, and phishing
Presentation	Dashboard, attack timeline, alert details, IOC view, technique mapping, agent report, and export

Table 4.1 Proposed system architecture layers

Fig. 4.1 shows the overall data flow: raw security data is preprocessed into behavioural features, which the hybrid detection engine and the rule engine both look at in parallel; the results are fused, correlated, enriched with IOC/ATT&CK context, and finally handed to the agentic layer, which is what actually produces the plan, report, and mitigation recommendation shown on the dashboard.

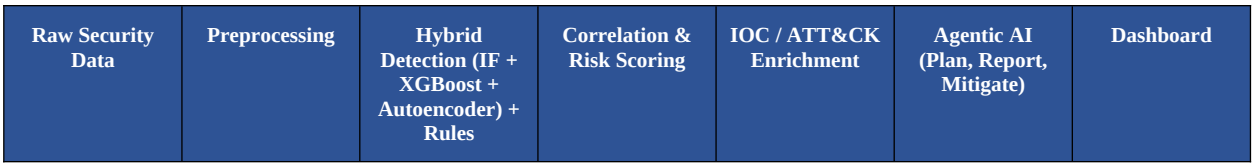


Fig. 4.1 High-level data flow of the proposed framework

4.2 Hybrid Machine-Learning Detection Component

Machine learning is not just a checkbox feature here — it is the core of the detection engine, and it is now a fused ensemble rather than one algorithm, so that known attacks, unknown behaviour, and cheap first-pass screening are each handled by whichever model is actually good at that job.

4.2.1 Isolation Forest — Unsupervised Baseline

Isolation Forest stays in as the fast, unsupervised baseline. It isolates points that are easy to separate from the rest of the data and gives an anomaly score cheaply, which is why it works well as an always-on first pass over incoming events [2].

4.2.2 XGBoost — Known-Attack Classification

A supervised XGBoost classifier is trained on the MITRE ATT&CK-labelled records in UWF-ZeekData24 to recognise known attack categories — credential access, reconnaissance, exfiltration, initial access, and so on — with the kind of precision reported for gradient-boosted classifiers on labelled network-intrusion data [8].

4.2.3 Deep Autoencoder — Unknown/Zero-Day Behaviour

A deep autoencoder is trained only on normal, baseline traffic. At inference time, anything with a high reconstruction error gets flagged as unusual — behaviour the rule engine and the supervised classifier were never trained to recognise — following the same class-imbalance and novel-attack reasoning behind recent autoencoder-based IDS work [8].

4.2.4 Score Fusion

The three scores — Isolation Forest's anomaly score, XGBoost's class probability, and the autoencoder's reconstruction error — are combined into one documented, fused risk score (a weighted combination or a small meta-classifier, depending on what works best in testing), which then goes to the rule engine for a second opinion and to the agentic layer as the main evidence for a case.

Step	Description
1. Data Collection	Collect UWF-ZeekData24 records and/or controlled lab-generated events.
2. Cleaning	Handle missing values, duplicate records, inconsistent timestamps, and irrelevant fields.
3. Feature Engineering	Aggregate events into time windows and create numerical behavioural features.
4. Model Fitting	Fit Isolation Forest, train XGBoost on labelled categories, and train the autoencoder on baseline traffic.
5. Scoring & Fusion	Generate per-model scores, fuse into a single risk score, and apply a documented threshold.
6. Evaluation	Measure detection quality per model and for the fused ensemble; analyse false positives/negatives.
7. Integration	Send fused results to the correlation engine, dashboard, and the agentic AI layer.

Table 4.2 Hybrid detection pipeline steps

4.3 Agentic AI Layer

As directed by our supervisor, this stays a single autonomous agent rather than a multi-agent setup — one agentic AI model handling report analysis, investigation planning, report generation, and mitigation recommendation. That keeps things realistic for a 12-week timeline while still automating the parts that matter, and avoids the coordination and evaluation headaches a full multi-agent SOC would bring [9].

4.3.1 Responsibilities

- Report/Log Analysis — reads correlated cases, fused risk scores, IOCs, and ATT&CK mappings produced by the earlier layers.
- Investigation Planning — proposes a step-by-step investigation plan for each case (what evidence to check next, which hosts/users are implicated, likely attack stage).
- Automated Report Generation — produces a human-readable incident-response report summarising the timeline, evidence, and ATT&CK techniques involved.
- Mitigation Recommendation — for common threat classes (ransomware, malware, phishing), recommends or, only inside the isolated lab and with human confirmation, triggers a predefined containment action (e.g., isolate a lab host, block an IOC in a test rule, draft a phishing-takedown notice).

4.3.2 Architecture and Guardrails

The agent is built as an LLM-based orchestrator with tool access to the correlation database, the IOC/ATT&CK store, and a small set of pre-approved actions. It is restricted to the isolated lab at every step — it can read case data and simulated system state, and can only carry out pre-approved, reversible, sandboxed actions, always with a human confirming before anything actually happens. It never acts on a real or production system on its own.

4.4 Dataset and Data Collection Strategy

The data strategy is still hybrid, just anchored on a more current dataset instead of an older benchmark.

Data Source	Purpose	Examples / Direction
UWF-ZeekData24 (primary)	Training, testing, and quantitative evaluation of the hybrid detection ensemble	Enterprise-scale, Zeek-based, MITRE ATT&CK-labelled network attack traffic (2025) [7]
Self-generated dataset (fallback)	Used only if UWF-ZeekData24 pre-processing/labelling proves impractical within the timeline	Controlled lab traffic captured and labelled by the team
Controlled lab	Generate realistic artefacts and demonstrate the full investigation + agentic workflow	Isolated VMs, benign simulations, controlled ransomware/malware/phishing scenarios, system logs, network captures
Forensic artefacts	Support timeline reconstruction and evidence correlation	File metadata, hashes, selected disk artefacts, memory-analysis outputs, and event records

Table 4.3 Data sources and their purpose

One thing worth being upfront about: a model trained on network-flow features is a network-anomaly detector, not something that catches every endpoint compromise, malware sample, or forensic artefact — we will not claim otherwise. The same goes for the agent's mitigation output: it is a recommendation validated in the lab, not a claim that the system can safely act on its own in production.

4.5 Functional Modules

Module	Key Functions
1. Data Ingestion	Import UWF-ZeekData24 records, logs, CSV/JSON, network-flow data, forensic outputs, and analysis reports.
2. Preprocessing & Feature Engineering	Parse, normalise, aggregate, clean, and transform data into model-ready features.
3. Hybrid Detection Engine	Fit/run Isolation Forest, XGBoost, and the autoencoder; fuse scores and apply thresholds.
4. Rule-Based Detection	Detect known patterns such as brute force, abnormal login behaviour, suspicious process activity, or traffic spikes.
5. Correlation & Case Management	Link events by host, user, IP, timestamp, IOC, and incident identifier.
6. Digital Forensics	Import selected artefact findings, hashes, timestamps, file information, and memory-analysis results.
7. Threat Intelligence	Extract IOCs, enrich selected indicators, and map behaviour to MITRE ATT&CK.
8. Agentic AI Module	Analyse the correlated case, plan the investigation, generate the report, and recommend mitigation for ransomware, malware, and phishing.
9. Dashboard & Reporting	Display alerts, fused risk scores, attack timeline, IOCs, evidence, techniques, agent plan/report, and exportable output.

Table 4.4 Functional modules of the framework

4.6 Proposed Tools and Technologies

Category	Proposed Technologies
Programming & ML	Python, pandas, NumPy, scikit-learn (Isolation Forest), XGBoost, TensorFlow/PyTorch (autoencoder), Jupyter Notebook
Agentic AI	An LLM API/model with tool-calling support, orchestrated with a lightweight agent framework (e.g., LangChain/LangGraph or an equivalent custom orchestrator)
Backend	FastAPI or Flask, REST APIs, Python-based processing services
Frontend	React or a lightweight Flask-based web interface
Database	PostgreSQL or SQLite for prototype storage; Elasticsearch/OpenSearch if needed
Digital Forensics	Autopsy, The Sleuth Kit, FTK Imager, Volatility 3
Malware Analysis	Ghidra or PEStudio, YARA, controlled sandbox reports
Network Analysis	Wireshark; UWF-ZeekData24 (Zeek-based) network-flow data

Threat Intelligence	MITRE ATT&CK, VirusTotal or AbuseIPDB where appropriate, OTX/MISP as optional integrations
Reporting	Matplotlib, dashboard charts, Markdown/LaTeX, PDF report generation

Table 4.5 Proposed tools and technologies

4.7 Detection Evaluation Plan

The hybrid detection engine is evaluated per model and as a fused ensemble.

- Precision, Recall, F1-score: computed per model (Isolation Forest, XGBoost, autoencoder) and for the fused score, where labels are available.
- False-positive rate: amount of normal activity incorrectly flagged as anomalous, per model and fused.
- Confusion matrix: used for the labelled XGBoost classification and for fused-score thresholding experiments.
- Runtime and resource usage: practical suitability for a student-scale security-monitoring system, including autoencoder training cost.
- Ablation comparison: rule-based detection alone vs. rule-based + single-model ML vs. rule-based + fused hybrid ensemble, to isolate the contribution of fusion.

4.8 Agentic AI Evaluation Plan

The agentic layer is evaluated separately from the detection engine, since a good detection score does not guarantee a useful plan or report.

- Task completion rate: proportion of cases for which the agent produces a complete plan, report, and mitigation recommendation.
- Groundedness/factual accuracy: whether the agent's report and plan are supported by the actual correlated evidence, checked against a human-analyst reference for a sample of cases.
- Plan quality: human-rated relevance and completeness of the proposed investigation steps.
- Mitigation appropriateness: whether the recommended containment action matches the identified threat class (ransomware, malware, phishing) and stays within the approved, sandboxed action set.
- Latency: end-to-end time from a fused detection to a produced agent report.

4.9 Demonstration Scenario

The final demonstration uses a controlled, isolated lab scenario that exercises the complete framework end to end, including at least one ransomware-style, one malware-style, and one phishing-style simulated scenario:

- Generate normal baseline activity in a lab environment.
- Introduce a controlled suspicious activity pattern (e.g., simulated ransomware file-encryption behaviour, a malware execution chain, or a phishing-triggered credential-access attempt).
- Collect relevant logs and forensic artefacts.
- Run preprocessing and hybrid ML scoring (Isolation Forest, XGBoost, autoencoder, fusion).
- Trigger rule-based detections and correlate them with fused ML output.
- Extract IOCs and map observed behaviour to MITRE ATT&CK.
- Hand the correlated case to the agentic AI layer to produce an investigation plan, an incident report, and a mitigation recommendation.
- Display the incident timeline, evidence, and agent output in the dashboard.

4.10 Scope and Safety Boundaries

In Scope

- Controlled lab-based incident simulation using owned or authorised virtual machines.
- UWF-ZeekData24 and other public cybersecurity datasets and publicly available analysis reports (used under their stated licences).
- Static analysis and carefully controlled analysis of suitable samples or benign test artefacts.
- Log parsing, hybrid feature engineering, fused anomaly/attack detection, alert correlation, IOC extraction, and ATT&CK mapping.
- A single agentic AI layer that reads correlated case data and recommends or triggers only pre-approved, sandboxed, reversible actions with human confirmation.
- A working dashboard, evaluation results, technical documentation, and a research-style report.

Out of Scope

- Executing unknown malware on personal, production, or internet-connected systems.
- Unauthorised access to third-party systems, networks, accounts, or data.
- Live incident response or autonomous mitigation on a real, ongoing attack outside the isolated lab.
- Building a complete commercial SIEM/SOAR, full reverse-engineering suite, or production-grade malware sandbox.
- Claiming universal detection performance, or fully autonomous production-safe mitigation, from a student-scale prototype.

4.11 Proposed 12-Week Timeline

Week(s)	Milestone
1–2	Literature review, architecture design, UWF-ZeekData24 acquisition/exploration, and isolated lab setup
3–4	Data preprocessing, incident simulation, log parsing, and database design
5–6	Feature engineering; Isolation Forest, XGBoost, and autoencoder implementation; score-fusion design and baseline evaluation
7	Rule-based detection, event correlation, and forensic artefact integration
8	IOC extraction, threat-intelligence enrichment, and MITRE ATT&CK mapping
9–10	Agentic AI layer: orchestrator, tool integration, investigation-planning and report-generation prompts, and sandboxed mitigation actions
11	Dashboard integration of detection, rules, forensics, CTI, and agent output; end-to-end testing
12	Evaluation, performance analysis, research-paper drafting, documentation, and submission

Table 4.6 Proposed project timeline

5. HARDWARE AND SOFTWARE REQUIREMENTS

5.1 Hardware Requirements

- A development workstation/laptop with a minimum quad-core processor and 16 GB RAM (recommended for running isolated VMs alongside autoencoder training and agent inference).
- At least 50–100 GB of free disk space for datasets, virtual machine images, logs, and model artefacts.
- A hypervisor-capable machine (VT-x/AMD-V enabled) to host isolated virtual machines for the controlled lab scenario.
- A GPU or cloud compute instance is recommended for autoencoder training and is strongly recommended if a locally hosted LLM is used for the agentic layer.

5.2 Software Requirements

- Operating System: Windows 10/11 or Ubuntu 20.04+ (host), with isolated Windows/Linux guest VMs for lab simulation.
- Programming: Python 3.10+, with pandas, NumPy, scikit-learn, XGBoost, TensorFlow/PyTorch, and Jupyter Notebook.
- Agentic AI: an LLM API/model with tool-calling support and an agent-orchestration library.
- Backend: FastAPI or Flask for REST APIs.
- Frontend: React, or a lightweight Flask-based web interface for the dashboard.
- Database: PostgreSQL or SQLite for prototype storage; Elasticsearch/OpenSearch if log-scale search is required.
- Digital Forensics Tools: Autopsy, The Sleuth Kit, FTK Imager, Volatility 3.
- Malware/Network Analysis: Ghidra or PEStudio, YARA, Wireshark (used only on controlled/benign or lab-generated captures).
- Threat Intelligence: MITRE ATT&CK reference data; VirusTotal or AbuseIPDB APIs where appropriate; OTX/MISP as optional integrations.
- Virtualisation: VirtualBox or VMware Workstation for the isolated lab environment.

6. EXPECTED OUTCOMES

6.1 Expected Outcomes

- A working, complete agentic AI-enhanced cybersecurity investigation framework.
- A documented hybrid detection pipeline (Isolation Forest, XGBoost, autoencoder, fusion) with feature engineering and per-model evaluation results.
- A rule-based and hybrid-ML-assisted detection workflow for suspicious activity, evaluated on the current UWF-ZeekData24 dataset.
- A unified incident timeline connecting logs, forensic artefacts, alerts, IOCs, and ATT&CK techniques.
- A single agentic AI layer that plans investigations, drafts incident reports, and recommends sandboxed mitigation for ransomware, malware, and phishing.
- A dashboard showing fused risk scores, alert severity, attack-chain context, evidence, and agent-generated output.
- A research-style evaluation discussing detection performance, agent reliability, limitations, and practical applicability.
- Demonstrable skills in machine learning, agentic AI/LLM orchestration, Python development, cybersecurity monitoring, digital forensics, and threat intelligence.

6.2 Potential Research Contribution

The contribution here is not a new ML algorithm or a new LLM — it is putting together and properly evaluating a complete framework that fuses a hybrid detection ensemble with a single autonomous agentic layer for investigation and mitigation. That combination is still fairly under-studied, sitting right at the intersection of ML-based detection work [4][8] and agentic SOC automation [9].

A suitable research question is:

"How effectively can a fused hybrid detection ensemble (Isolation Forest, XGBoost, and a deep autoencoder), combined with a single agentic AI layer, support investigation planning, reporting, and mitigation of ransomware, malware, and phishing activity, relative to rule-based detection and dashboard-only reporting?"

What we actually want to find out is whether the fused ensemble beats any single model on coverage and prioritisation, and whether the agent's plans, reports, and mitigation suggestions are accurate enough, grounded enough in the real evidence, and useful enough to actually save an analyst time. We are hoping to write this up as a short paper for a UG research symposium or workshop track, alongside the software itself.

7. CONCLUSION

This project sets out to build a practical, research-flavoured cybersecurity framework that ties together hybrid machine learning, digital forensics, threat intelligence, and a single agentic AI layer. By fusing Isolation Forest, XGBoost, and a deep autoencoder into one detection engine, testing on the current UWF-ZeekData24 dataset, and connecting the resulting cases to an agent that plans the investigation, writes the report, and suggests mitigation for ransomware, malware, and phishing, the project moves past being just a detection demo into something closer to a complete, end-to-end framework.

The plan is to build a focused, well-evaluated prototype rather than trying to rebuild every feature of a commercial SIEM or SOAR product, and to keep the agentic layer to one well-scoped orchestrator instead of an unmanaged multi-agent system. Done properly — with a solid evaluation plan for both the detection ensemble and the agent — this should make for a strong major-project result and a reasonable basis for a short research paper too.

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